

Speech Synthesis

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Developing TTS Systems

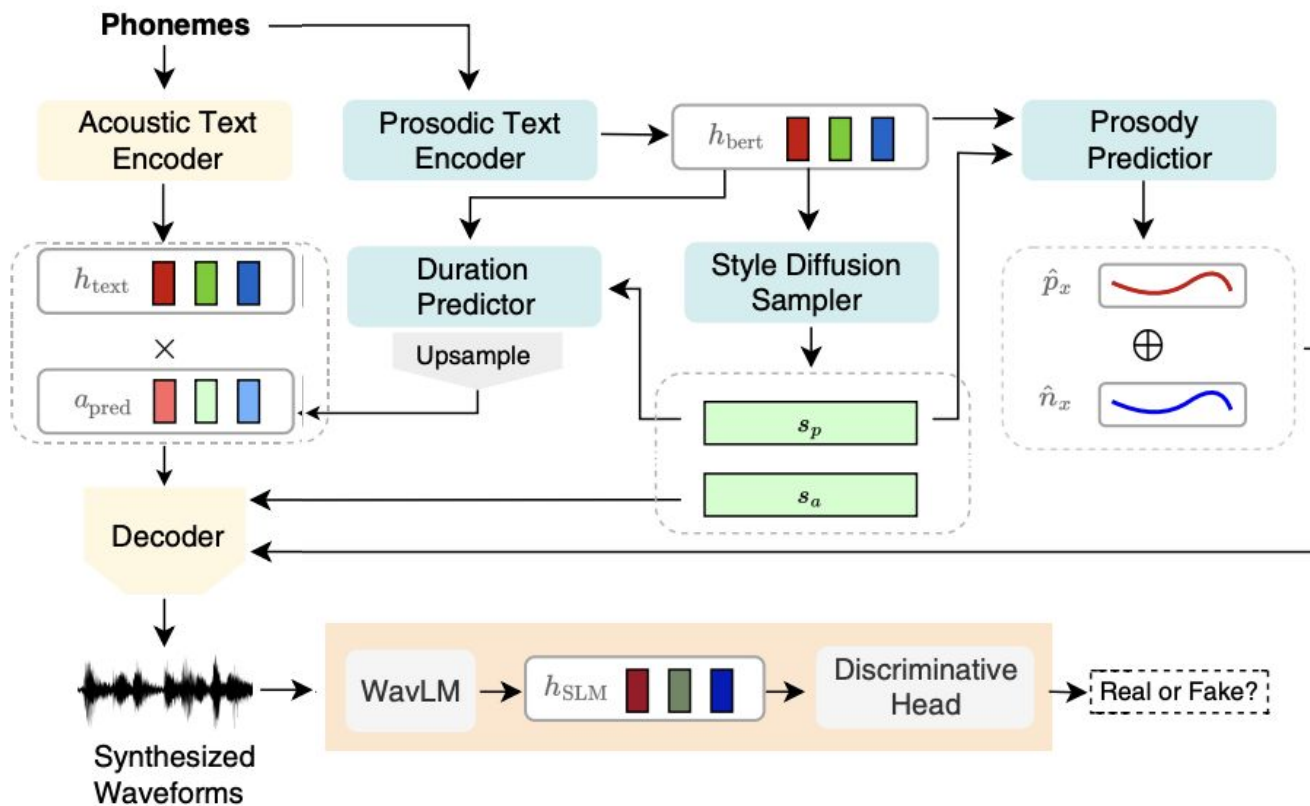
Summary of Previous class

- Converts text to acoustic feature (Mel-spectrogram)
 - Tacotron 2
 - Transformer TTS
- Acoustic feature to waveform
 - Wavenet
 - Parallel WaveGAN
- E2E model
 - StyleTTS2

StyleTTS2

- StyleTTS 2 is a state-of-the-art open source text-to-speech model
- Non-autoregressive architecture (generates audio in parallel)
- Combines style diffusion and adversarial training
- Supports zero-shot voice cloning and style transfer
- Trained with Single Speaker (24 hrs), multi-speaker (245 hours of audio from 1,151 speakers)
- Model parameters ~108 M, latency-> RTF: 0.0185

StyleTTS2: Block diagram



Demo

<https://styletts2.github.io/>

TTS Development

Data Collection Planning:

- Define target speaker profile (age, gender, accent, speaking style)
- Decide on recording duration (typically 20 hours for good quality if you are starting from scratch, 2 hrs if you are planning to do just to fine-tuning)
- Define script content type (news, conversation, narrative, etc.)

Speaker Selection:

- Professional voice talent vs native speaker
- Clear pronunciation and consistent speaking style
- Voice quality assessment (recording sample test)

Text Script Preparation:

- Cover all phonemes and their combinations,
- different sentence types (questions, statements, commands)
- Include numbers, dates, currency amounts
- Balance sentence lengths
- Include domain-specific terminology if required

TTS Development: Recording Guidelines

- Room acoustics (low reverberation)
- Microphone selection and placement
- Consistent distance from microphone
- Background noise control
- Recording format (48kHz, 24-bit recommended)
- Regular breaks to maintain voice quality
- Monitor recording levels

TTS Development: Data filtering

- Audio Quality Issues:

- Background noise level
- Clipping/distortion
- Room reverberations
- Microphone pops/clicks
- Inconsistent volume levels

- Speech Quality Issues:

- Mispronunciations or Unclear articulation
- Breathing noises
- Mouth clicks
- Speed inconsistencies
- Emotional inconsistencies
- Hesitations/fillers

TTS Development: Data processing

- Sentence segmentation
- Silence removal at boundaries
- Volume normalization
- Noise reduction if needed
- Check for DC offset
- Sample rate consistency
- Bit depth standardization

Inference

https://colab.research.google.com/github/nvadigauvce/StyleTTS2/blob/main/Colab/StyleTTS2_Demo_LibriTTS_v2.ipynb

More samples:

https://colab.research.google.com/github/nvadigauvce/StyleTTS2/blob/main/Colab/StyleTTS2_Demo_LibriTTS.ipynb

Training:

https://colab.research.google.com/github/nvadigauvce/StyleTTS2/blob/main/Colab/StyleTTS2_Finetune_Demo.ipynb

Pre-trained model: Libri Speech trained with 245 hours of audio from 1,151 speakers.

Task: Fine-tuning with around 186 samples (~19 min) from LJSpeech dataset

References

- Klatt; "The Klatt-Talk text-to-speech system," in Proc. ICASSP, '82.
- Hunt, A. and Black, A. (1996). Unit selection in a concatenative speech synthesis system using a large speech database, in Proc. ICASSP 96, pp 373-376, Atlanta, Georgia.
- K. Tokuda, Y. Nankaku, T. Toda, H. Zen, J. Yamagishi, and K. Oura, "Speech synthesis based on hidden markov models", Proceedings of the IEEE, vol. 101-5, pp. 12341252, 2013.
- H. Zen, K. Tokuda, and A. W. Black, Statistical parametric speech synthesis, Speech Communication, vol.51(11), pp. 10391064, 2009
- Simon King, An introduction to statistical parametric speech synthesis, Sadhana Vol. 36, Part 5, October 2011, pp. 837{852
- Tokuda, K., Yoshimura, T., Masuko, T., Kobayashi, T., Kitamura, T. Speech parameter generation algorithms for HMM-based speech synthesis" in proceedings of ICASSP 2000.

References

- Xie, Tianxin, et al. "Towards Controllable Speech Synthesis in the Era of Large Language Models: A Survey." *arXiv preprint arXiv:2412.06602* (2024).
- X. Tan, T. Qin, F. Soong, and T.-Y. Liu, "A survey on neural speech synthesis," arXiv preprint arXiv:2106.15561, 2021
- Triantafyllopoulos, Andreas, et al. "An overview of affective speech synthesis and conversion in the deep learning era." *Proceedings of the IEEE* 111.10 (2023): 1355-1381.
- Zhang, Chenshuang, et al. "A survey on audio diffusion models: Text to speech synthesis and enhancement in generative ai." *arXiv preprint arXiv:2303.13336* (2023).
- van den Oord, Aaron; Dieleman, Sander; Zen, Heiga; Simonyan, Karen; Vinyals, Oriol; Graves, Alex; Kalchbrenner, Nal; Senior, Andrew; Kavukcuoglu, Koray, WaveNet: A Generative Model for Raw Audio, arXiv:1609.03499, 2016
- Adiga, Nagaraj, Vassilis Tsias, and Yannis Stylianou. "On the use of WaveNet as a statistical vocoder." *Proc. ICASSP*. IEEE, 2018.
- Shen, Jonathan, et al. "Natural tts synthesis by conditioning wavenet on mel spectrogram predictions." IEEE, *Proc. ICASSP*, 2018.
- Yamamoto, Ryuichi, Eunwoo Song, and Jae-Min Kim. "Parallel WaveGAN: A fast waveform generation model based on generative adversarial networks with multi-resolution spectrogram." *Proc. ICASSP 2020*
- Li, Yinghao Aaron, et al. "Styletts 2: Towards human-level text-to-speech through style diffusion and adversarial training with large speech language models." *Advances in Neural Information Processing Systems* 36 (2023): 19594-19621.

References

- ISCA Synthesis Interest Group: https://synsig.org/index.php/Main_Page (look there for the SPCC videos)
- CSTR, Simon Kings' page on speech synthesis: <http://www.speech.zone/courses/speech-synthesis/>
- CMU Festvox: <http://festvox.org/>
- HTS web page <http://hts.sp.nitech.ac.jp/> for more information on HMM based speech synthesis system (HTS)
- Google: <https://google.github.io/tacotron/>
- Espnet toolkit link: <https://github.com/espnet/espnet>

Thank You